

0 5 . 1

Describe how organisms are grouped in a phylogenetic classification system.

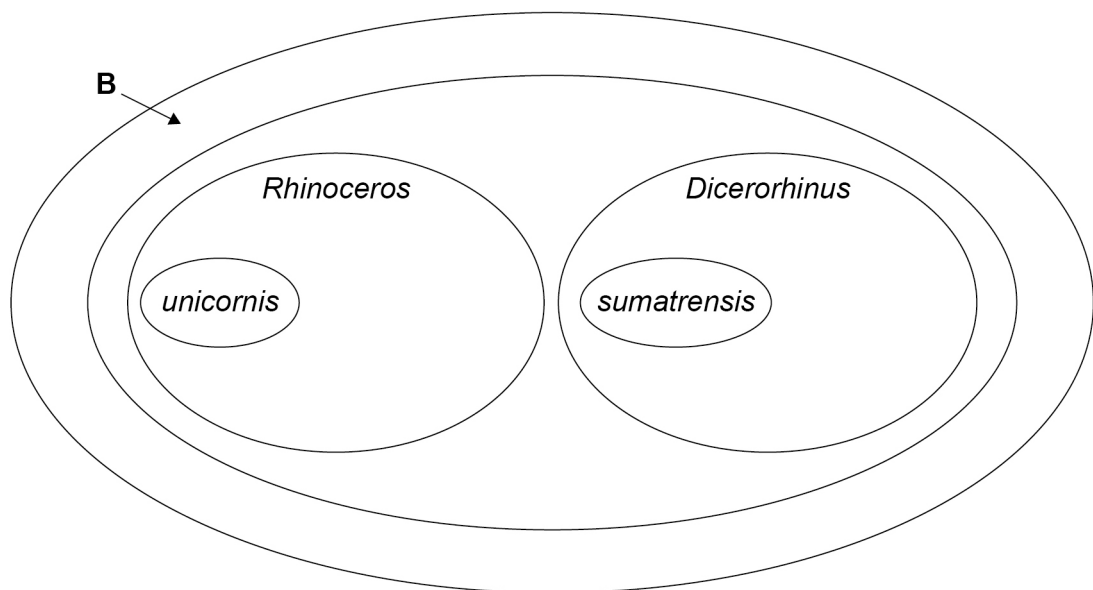
[2 marks]

Table 1 shows some of the taxa in the phylogenetic classification of a rhinoceros species.

Table 1

Taxon name	Scientific name
Class	Mammalia
Order	Perissodactyla
Family	Rhinocerotidae
Genus	<i>Rhinoceros</i>
Species	<i>unicornis</i>

Figure 4 shows the relationship between the taxa in the classification of two rhinoceros species: *Rhinoceros unicornis* and *Dicerorhinus sumatrensis*.

Figure 4

0 5 . 2

Use information in **Table 1** to give the **scientific** name of the taxon labelled **B** in **Figure 4**.

[1 mark]

0 5 . 3

Draw an oval on **Figure 4** to show the species *Rhinoceros sondaicus*.

[1 mark]

Question 5 continues on the next page

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The scientists:

- determined the DNA base sequence of the *cyt b* gene of each rhinoceros
- compared each *cyt b* DNA base sequence with that of **one** Indian rhinoceros (called the reference rhinoceros)
- calculated the percentage difference between each *cyt b* DNA base sequence and that of the reference rhinoceros.

Table 2

Investigated species of rhinoceros	Percentage difference in DNA base sequences compared with the reference Indian rhinoceros
Indian	2
Javan	5
Sumatran	13
White	14
Black	14

What can you conclude about the likely phylogenetic relationships between these species? Evaluate your conclusion.

[4 marks]

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0 5 . 5

A scientist obtained a rhinoceros horn confiscated from poachers and wanted to identify the species of rhinoceros that was killed for its horn.

He used the procedure described in Question **05.4** and calculated the difference in *cyt b* DNA as 14%.

What can you conclude from this result? Explain your answer.

Suggest a change to the procedure that will more precisely identify the rhinoceros species that provided the horn.

[2 marks]

Conclusion and explanation _____

Suggested change to the procedure _____

10

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